How LLMs Slay the Spire 2: Exploring the Boundary of LLMs in a Deckbuilding Roguelike Game

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Figure 1: Overview of STSAUTOBOT: LLMs play *Slay the Spire 2* as autonomous agents, making sequential decisions across combat, deck building, and map navigation.

Abstract

We present STSAUTOBOT, a framework for evaluating large language models (LLMs) as autonomous strategic agents in *Slay the Spire 2*, a procedurally generated roguelike deck-building game that demands long-horizon planning, real-time resource management, and adaptive combat tactics. We design a multi-dimensional benchmark comprising four performance indices Survival, Combat IQ, Deck Mastery, and Resource Management. Each scored on a 0-100 scale and aggregated into a weighted composite. We evaluate seven frontier LLMs (DeepSeek-V3.2, GPT-5.4, MiniMax-2.7, Kimi-K2.5, GLM-5, Gemini-3.1pro, and Opus-4.6) as the decision-making backbone of the agent. No model achieves a full victory, exposing a significant gap between current LLM capabilities and expert human play. We find that resource management and floor progression are strongly correlated, and that rest site strategy (heal vs. smith) varies significantly across models. Our benchmark (planned to be continuously updated) will be publicly available to support future research.[1]

[1] GNU Affero General Public License v3.0 only (AGPL-3.0-only). We plan to provide weekly updated results across multiple official daily runs distributed by the STS2 developers.

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across diverse tasks, from code generation to mathematical reasoning (Brown et al., 2020; Wei et al., 2022). A growing body of work explores their potential as autonomous agents in interactive environments, including video games (Wang et al., 2023; Wu et al., 2023), robotic control (Liang et al., 2022), and planning tasks (Huang et al., 2022). However, evaluating LLMs in domains that require *long-horizon strategic reasoning* under uncertainty remains an open challenge.

Slay the Spire 2 (STS2) is a single-player roguelike deckbuilding game that presents a compelling testbed for LLM strategic reasoning. Each run requires the agent to navigate a procedurally generated map across multiple acts, build a synergistic card deck through sequential decisions, manage health and gold resources, and execute turn-based combat against diverse enemies with hidden intent patterns. Unlike board games with perfect information or text adventures with limited action spaces, STS2 demands the integration of short-term tactical decisions (combat card play) with long-term strategic planning (deck construction, pathing, resource allocation) across hundreds of sequential decision points.

In this paper, we introduce STSAUTOBOT, a modular framework that connects LLMs to STS2 as the autonomous decision-making engine. We design a comprehensive **four-index benchmark** that evaluates agent performance along orthogonal dimensions:

1. **Survival Score** (30%): Floor progression and HP management
2. **Combat IQ** (30%): Turn efficiency and damage avoidance
3. **Deck Mastery** (20%): Deck building and curation quality
4. **Resource Management** (20%): Gold, relic, and potion utilization

We evaluate seven frontier LLMs as the agent backbone and find significant performance differences. Our main contributions are:

- A modular agent framework (STSAUTOBOT) connecting LLMs to STS2 with structured prompting and action validation.
- A multi-dimensional benchmark with four interpretable performance indices.
- An empirical evaluation of seven LLMs revealing that Gemini-3.1pro achieves the best overall performance (59.8/100, floor 33), while all models fail to achieve a full game victory.

Performance Index	Weight	Components
Survival Score	30%	3
Combat IQ	30%	4
Deck Mastery	20%	5
Resource Management	20%	4

Table 1: Benchmark index structure. Each index produces a score in $[0, 100]$, combined via weighted sum.

2 Method

2.1 System Architecture

STSAUTOBOT consists of six modules organized in a forward return pipeline (Figure 2):

1. **Game Client**[2]: Communicates with STS2 via MCP/HTTP, receiving raw game state and dispatching validated actions.
2. **State Enrichment**: Resolves card template variables, attaches relic/potion descriptions, and computes derived metrics (e.g., incoming damage, unblocked damage).
3. **Prompt Builder**: Converts enriched state into structured natural language prompts with available actions.
4. **LLM API**: Routes prompts to the LLM backend and receives JSON-formatted responses.
5. **JSON Parser + Validator**: Parses the LLM response, validates the action against the current game state, and handles retries on parse errors or timeouts.
6. **Action Executor**: Translates validated decisions into game commands sent back through the Game Client.

All decisions, game states, and LLM interactions are logged by the **State Logger** for post-hoc analysis and benchmarking.

2.2 Benchmark Design

We define four performance indices, each scored on a 0–100 scale and combined into a weighted composite:

$$S_{\text{overall}} = \sum_{i=1}^{4} w_i \cdot S_i \tag{1}$$

Table 1 summarizes the index weights and component counts.

[2] https://github.com/CharTyr/STS2-Agent

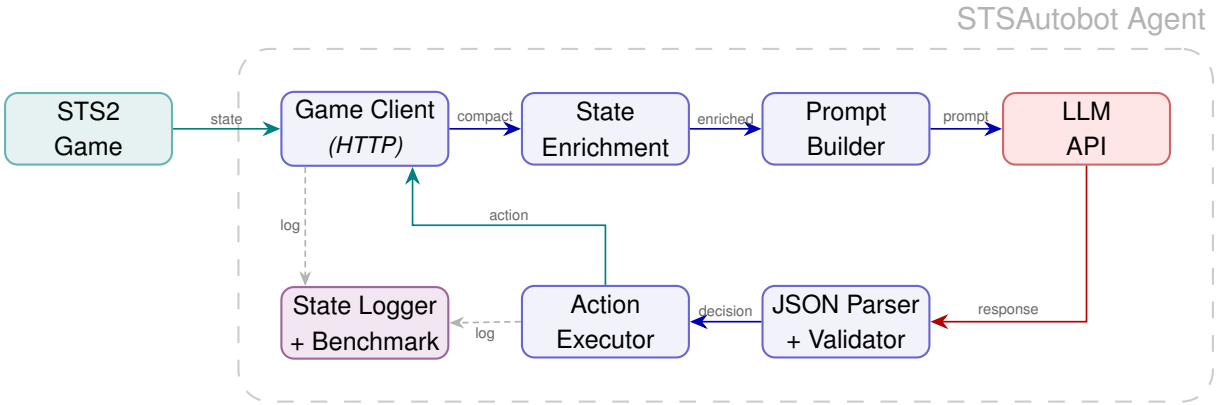

Figure 2: Architecture of the STSAUTOBOT framework. The top row shows the forward pipeline (game state → LLM); the bottom row shows the return pipeline (LLM response → action execution). Dashed gray arrows denote logging. The dashed bounding box delineates the agent boundary.

Index 1: Survival Score ($w_1 = 0.25$). Measures longevity and health management (Table 2):

$$S_f = \min(f/55,\ 1) \times 80 \tag{2}$$
$$S_r = q_r \times 20 \tag{3}$$

where f = final floor and $q_r \in [0, 1]$ = fraction of correct rest decisions (heal when HP $< 45\%$, smith when HP $\geq 60\%$).

Index 2: Combat IQ ($w_2 = 0.25$). Evaluates combat effectiveness at the floor and encounter level (Table 3):

$$S_{\text{fe}} = \max\big(0,\ 1 - (\bar{t}_f - 3)/12\big) \times 30 \tag{4}$$
$$S_{\text{cf}} = (n_{\text{clean}}/N_f) \times 30 \tag{5}$$
$$S_{\text{hl}} = \max\big(0,\ 1 - \bar{d}_f/15\big) \times 25 \tag{6}$$
$$S_{\text{cp}} = \mathcal{N}(\bar{c};\ 2.8,\ 1.5) \times 15 \tag{7}$$

where $\bar{t}_f$ = avg turns per combat floor, n_{clean} = floors with zero HP loss, $\bar{d}_f$ = avg HP lost per floor, and $\mathcal{N}$ is a Gaussian bell curve on avg cards per turn.

Index 3: Deck Mastery ($w_3 = 0.20$). Assesses deck building quality:

$$S_{\text{size}} = \mathcal{N}(|D|;\ 20,\ 5) \times 25 \tag{8}$$
$$S_{\text{smith}} = \min(n_s/3,\ 1) \times 25 \tag{9}$$
$$S_{\text{rem}} = \min(n_r/4,\ 1) \times 20 \tag{10}$$
$$S_{\text{net}} = \max(0,\ 1 - \max(0,\ n_a - n_r - 10)/10) \times 15 \tag{11}$$
$$S_{\text{cur}} = \min((n_r + n_s)/f \cdot 10/3,\ 1) \times 15 \tag{12}$$

where $|D|$ = final deck size, n_s = cards smithed at rest sites, n_r = cards removed, n_a = cards added, and f = final floor. Net Growth Control penalizes net deck growth exceeding 10. Curation Rate measures active deck management (removals + smiths) per 10 floors.

Index 4: Resource Management ($w_4 = 0.10$). Tracks four resource dimensions (Table 4):

$$S_{\text{gold}} = \min(G_s/(G_0 + G_e),\ 1) \times 30 \tag{13}$$
$$S_{\text{relic}} = \min(n_r/(f/10 \cdot 3),\ 1) \times 30 \tag{14}$$
$$S_{\text{pot}} = \min(p/8,\ 1) \times 20 \tag{15}$$
$$S_{\text{shop}} = \min(a_s/(2 \cdot v_s),\ 1) \times 20 \tag{16}$$

where G_s = gold spent, G_0 = initial gold, G_e = gold earned, p = potions used (from state snapshots), a_s = shop actions, v_s = shop visits.

3 Experiments

3.1 Setup

All experiments are conducted on the **March 26, 2026 Daily Challenge** of STS2, which features the **Silent** character class at **Hard level 4**. The daily challenge provides a fixed seed, ensuring all models face identical procedural generation (same map layout, card offerings, enemy sequences, and event options). Each model is given a single complete run from floor 1 until death or victory. The agent communicates with the game via WebSocket and receives the full observable game state at each decision point. LLM prompts include the current screen type (combat, map, shop, event), available actions with descriptions, and relevant context (HP, gold, hand, enemies, relics). Models output structured JSON responses with an action key and reasoning explanation.

	DeepSeek-V3.2	GPT-5.4	MiniMax-2.7	Kimi-K2.5	GLM-5	Gemini-3.1pro	Opus-4.6
Floor Prog. (80)	10.2	24.7	16.0	40.7	40.7	**48.0**	**48.0**
Rest Quality (20)	0.0	**20.0**	**20.0**	**20.0**	**20.0**	**20.0**	12.0
Total (100)	10.2	44.7	36.0	60.7	60.7	**68.0**	60.0

Table 2: Survival Score breakdown. Floor Progression dominates (80 pts), directly rewarding deeper runs.

	DeepSeek-V3.2	GPT-5.4	MiniMax-2.7	Kimi-K2.5	GLM-5	Gemini-3.1pro	Opus-4.6
Floor Eff. (30)	24.2	6.7	23.0	19.3	10.4	7.5	**22.7**
Clean Floor (30)	5.0	9.0	0.0	8.6	4.6	8.8	**10.0**
HP Loss Eff. (25)	**9.4**	6.3	4.3	5.2	8.6	4.3	2.9
Card Play (15)	13.0	14.9	**15.0**	10.3	14.2	14.2	**15.0**
Total (100)	**51.6**	37.0	42.3	43.4	37.8	34.8	50.5

Table 3: Combat IQ breakdown. Floor Efficiency measures avg turns per combat floor (fewer = better). Clean Floor and HP Loss Efficiency are per-floor. Potion usage is scored under Resource Management to avoid overlap.

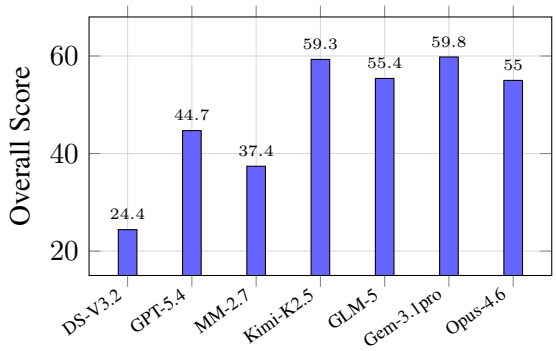

Figure 3: Overall benchmark scores across the seven evaluated LLMs.

3.2 Models

We evaluate seven LLMs as the agent's decision backbone, as summarized in Table 5. All models are accessed through an OpenAI-compatible API endpoint with a temperature of 0.2, timeout of 20 seconds, and up to 2 retries on failure.

4 Results

4.1 Overall Performance

Table 6 presents the benchmark scores across all seven models. Gemini-3.1pro achieves the highest overall score (59.8), closely followed by Kimi-K2.5 (59.3) and GLM-5 (55.4). Opus-4.6 (55.0) and GPT-5.4 (44.7) form a middle tier. DeepSeek-V3.2 trails at 24.4 due to early elimination at floor 7.

4.2 Dimension-Level Analysis

Figure 4 shows the four-index profile for each model, revealing distinct strategic signatures.

Survival. Gemini-3.1pro leads (68.0) by reaching floor 33, followed by Kimi-K2.5 and GLM-5 (both 60.7) at floor 28. DeepSeek-V3.2's poor survival (10.2) stems from early elimination at floor 7 with a 100% combat rate it visited only combat floors before dying.

Combat IQ. DeepSeek-V3.2 achieves the highest Combat IQ (51.6) with strong floor efficiency, followed by Opus-4.6 (50.5). MiniMax-2.7 completed zero clean floors despite decent floor efficiency (42.3), while Opus-4.6 achieved the most clean floors (5 out of 15 combat floors).

Deck Mastery. Gemini-3.1pro leads (75.2) with 4 smiths across 7 rest visits and 7 card removals, demonstrating the most active deck curation (3.3 actions per 10 floors). Kimi-K2.5 follows at 69.1 with the highest removal count (6). Opus-4.6 scores only 42.6 due to zero smithing across 5 rest visits, prioritizing healing over deck improvement.

Resource Management. GLM-5 leads (76.3) with the highest gold utilization (94.7%) and strong relic acquisition. Kimi-K2.5 follows at 71.2 with the most productive shop visits. MiniMax-2.7 scores only 25.7 with zero gold spent across 256 available, never engaging with the shop system.

4.3 Combat and Deck Statistics

Table 7 presents detailed combat and deck building metrics.

Gemini-3.1pro and Opus-4.6 both reach floor 33, the deepest of all models. Gemini-3.1pro accumulates 204 combat turns with an HP loss of 12.4 per combat floor, while DeepSeek-V3.2 achieves the lowest HP loss per floor (9.3) but is eliminated too

	DeepSeek-V3.2	GPT-5.4	MiniMax-2.7	Kimi-K2.5	GLM-5	Gemini-3.1pro	Opus-4.6
Gold Util. (30)	3.9	23.0	0.0	27.9	**28.4**	19.7	18.5
Relic Acq. (30)	0.0	5.9	18.2	10.7	17.9	**27.3**	18.2
Potion Use (20)	2.5	7.5	7.5	12.5	15.0	7.5	**20.0**
Shop Prod. (20)	0.0	0.0	0.0	**20.0**	15.0	15.0	10.0
Total (100)	6.4	36.3	25.7	71.2	**76.3**	69.5	66.7

Table 4: Resource Management breakdown. Gold Utilization is computed as gold spent / (initial gold + gold earned). Relic Acquisition is scaled by relics per 10 floors. Potion Use is verified from state snapshots. Shop Productivity measures buy/remove actions per shop visit.

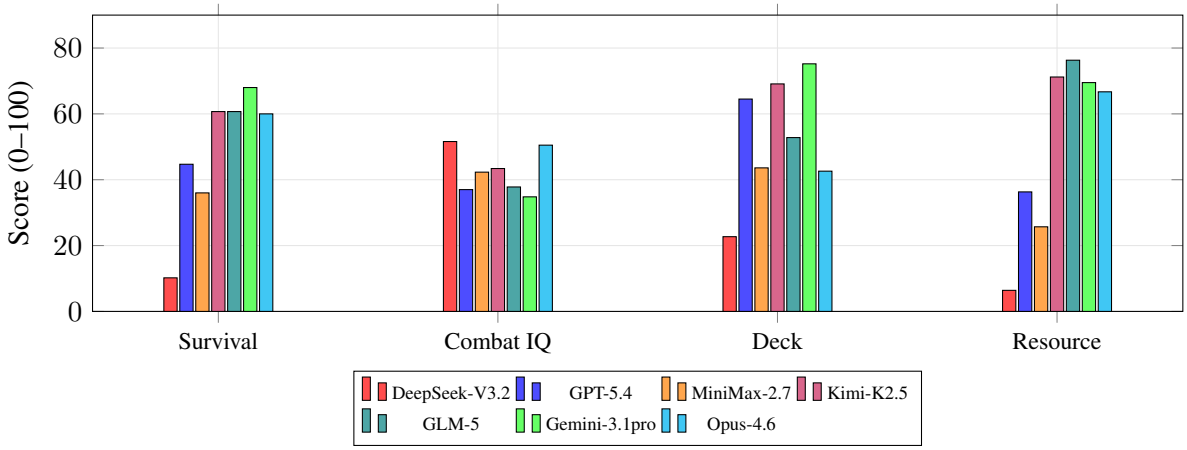

Figure 4: Four-index performance profiles across seven models. Each cluster shows all models' scores for one benchmark dimension.

Model	Total Dec.	Potions	Relics
DeepSeek-V3.2	141	1	0
GPT-5.4	351	3	1
MiniMax-2.7	124	3	2
Kimi-K2.5	494	5	3
GLM-5	505	6	5
Gemini-3.1pro	676	3	9
Opus-4.6	511	11	6

Table 5: Model run statistics. Total Dec. = total decisions made during the run. All runs use the Silent Hunter at Ascension 0.

early to benefit. Opus-4.6 has the lowest combat rate (45.5%), suggesting a more diversified pathing strategy through shops, events, and rest sites.

5 Discussion

Resource Management Drives Progression. The strongest correlation in our results is between Resource Management and final floor reached. The top four models by floor progression, Gemini-3.1pro (floor 33), Opus-4.6 (floor 33), GLM-5 (floor 28), and Kimi-K2.5 (floor 28) all score above 66 on Resource Management. In contrast, MiniMax-2.7 spent zero gold despite earning 157, and DeepSeek-V3.2 spent only 20 of 55 earned, both corresponding to shorter runs. This suggests that LLMs' ability to engage with economic subsystems is a key differentiator in complex game environments.

Tactical vs. Strategic Reasoning. DeepSeek-V3.2 achieves the highest Combat IQ (51.6) but the lowest overall score (24.4), as its strong per-floor efficiency cannot compensate for early elimination at floor 7. Conversely, Kimi-K2.5 has moderate Combat IQ (43.4) but excels at resource management (71.2) and deck mastery (69.1), achieving the second-highest overall score (59.3). This tactical-strategic gap may reflect differences in how models handle local versus global optimization.

Rest Site Strategy: Heal vs. Smith. Models diverge sharply on the heal-vs-smith tradeoff at rest sites. MiniMax-2.7 smithed at its only rest visit (100%), while Opus-4.6 healed at all 5 visits (0% smith rate). GPT-5.4 and GLM-5 both smithed at 2 of 3 visits (67%), and Gemini-3.1pro smithed 4 of 7 (57%). Kimi-K2.5 split evenly with 2 smiths and 2 heals across 4 visits. Opus-4.6's exclusive preference for healing may explain its higher total

	DeepSeek-V3.2	GPT-5.4	MiniMax-2.7	Kimi-K2.5	GLM-5	Gemini-3.1pro	Opus-4.6
Survival Score (30%)	10.2	44.7	36.0	60.7	60.7	**68.0**	60.0
Combat IQ (30%)	**51.6**	37.0	42.3	43.4	37.8	34.8	50.5
Deck Mastery (20%)	22.7	64.5	43.6	69.1	52.8	**75.2**	42.6
Resource Mgmt (20%)	6.4	36.3	25.7	71.2	**76.3**	69.5	66.7
Overall	24.4	44.7	37.4	59.3	55.4	**59.8**	55.0

Table 6: Main benchmark results across seven LLMs. Each index is scored on $[0, 100]$ with weights shown in parentheses. Bold indicates best per row.

	DeepSeek-V3.2	GPT-5.4	MiniMax-2.7	Kimi-K2.5	GLM-5	Gemini-3.1pro	Opus-4.6
Final Floor	7	17	11	28	28	**33**	**33**
Combat Turns	32	123	29	102	141	**204**	89
Combat Rate %	**100.0**	62.5	50.0	51.9	48.1	51.5	45.5
Total HP Lost	56	112	**62**	166	128	211	256
HP Lost/Floor	**9.3**	11.2	12.4	11.9	9.8	12.4	13.3
Clean Floors	1	3	0	4	2	**5**	**5**
Gold Earned	55	182	157	383	339	**480**	393
Gold Spent	20	215	0	**449**	415	380	303
Final Deck Size	15	18	**14**	24	26	25	22
Cards Removed	0	2	1	6	3	**7**	3
Cards Upgraded	0	2	1	2	2	**4**	0

Table 7: Combat and deck building statistics across seven models. Bold indicates best per row.

HP lost per floor (13.3) despite reaching floor 33 it prioritized short-term survival over long-term deck quality (Shinn et al., 2023).

No Model Achieves Victory. All seven models died during their runs, with the best (Gemini-3.1pro and Opus-4.6) reaching floor 33 of approximately 55 floors. This 60% completion rate highlights the substantial gap between current LLM capabilities and expert human play, where experienced players achieve victory rates above 50% even at high ascension levels. The game's requirement for emergent strategy adapting deck building and pathing to specific card and relic synergies, remains beyond current zero-shot LLM reasoning.

Future Directions. Several avenues could improve LLM performance: (1) multi-turn reflection (Shinn et al., 2023; Yao et al., 2023) to enable self-correction, (2) game-specific knowledge retrieval augmenting the prompt with card synergy databases, (3) hybrid approaches combining LLM strategic reasoning with search-based tactical combat, and (4) fine-tuning on expert gameplay traces.

Limitations

Our study has several limitations. First, each model is evaluated on a **single run**, which limits statistical significance. Run-to-run variance due to procedural generation (different maps, card offerings, enemy sequences) could substantially affect results. Future work should evaluate each model across multiple runs. Second, we lack a **human expert baseline**. While we know that expert players achieve high win rates, we do not have a controlled human run under identical conditions for direct comparison.

Ethical Considerations

Our research evaluates LLMs as autonomous agents within a single-player video game (*Slay the Spire 2*) and raises no direct risks to human subjects. We nonetheless address several ethical dimensions relevant to this work.

Nature of the Task. The LLMs in our study are tasked exclusively with playing a commercial deck-building roguelike game. The game involves strategic card combat against fictional, procedurally generated enemies there are no depictions of real-world violence, hate, or harm. All "combat" is turn-based and abstract, consisting of playing numbered cards to deal damage or gain block against fantasy creatures. The agent's goal is survival and progression through the game's floors, not causing harm in any real-world or simulated-realistic sense.

Full Game State Disclosure. Each LLM receives the complete observable game state at every decision point: current HP, gold, the full card hand

with descriptions, all enemy intents (including upcoming damage values), deck composition, relics, potions, map topology, and available actions. The agent is never asked to deceive, manipulate, or exploit any entity human or otherwise. It simply selects the best available action (e.g., "play card 3" or "choose map node 2") and provides a natural language explanation of its reasoning. The structured JSON output format further constrains the LLM to well-defined action spaces with no opportunity for open-ended harmful generation.

No Deceptive or Adversarial Use. Our framework does not train, fine-tune, or reinforce any model. All evaluations are zero-shot: the LLM receives a prompt describing the game state and responds with a single action. No model weights are modified, and no reward signals are used to shape behavior. The benchmark measures existing capabilities rather than cultivating new ones.

Commercial Game Content. Our framework interacts with *Slay the Spire 2* through its modding communication interface. We do not redistribute game assets, reverse-engineer proprietary code, or circumvent access controls. All card descriptions and game mechanics referenced in this paper are used solely for research purposes under fair use.

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